

**DOES STUDY TIME PREDICT TEST PERFORMANCE AMONG DORAL
ACADEMY STUDENTS?**

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AP Statistics Research Project
Extended Competition Version

I. Introduction

Academic success is influenced by numerous variables, including motivation, attendance, socioeconomic status, learning style, classroom engagement, sleep quality, and study habits. Among these variables, study time has often been considered one of the strongest predictors of academic success. Students, parents, and educators commonly assume that more study time produces better grades; however, educational research suggests the relationship is more complex than simple quantity of time.

Nonis and Hudson (2010) found positive associations between study time and academic outcomes, though they noted diminishing returns as study hours increased. Plant et al. (2005) argued that the quality and structure of studying may be more important than raw study duration. Their findings suggested deliberate practice and active engagement produce stronger results than passive review. Additionally, Credé and Kuncel (2008) conducted a meta-analysis and concluded that study habits significantly predict examination performance and GPA.

Despite extensive literature, most studies focus on college populations. High school students experience different academic environments, schedules, and motivational pressures. Doral Academy students participate in rigorous coursework and standardized testing environments. Understanding how study time influences performance within this population may provide practical implications for student success.

II. Statistical Question and Data Collection

Research Question: Does the number of hours students spend studying for a major test significantly predict test performance among Doral Academy students?

Null Hypothesis: $H_0: \mu=5$ hours

Alternative Hypothesis: $H_a: \mu \neq 5$ hours

Population: All Doral Academy High School students grades 9–12

Sampling frame: Students currently enrolled grades 9–12

Sample size: $n=50$

To ensure representation across grade levels, stratified random sampling was used. Approximately 12–13 students per grade participated. Students were recruited during study hall periods and invited to participate in anonymous academic surveys.

The survey included:

- Hours studied
- Test scores
- Grade level
- Gender demographics

Explanatory Variable (x): Hours studied (0–12)

Response Variable (y): Test score percentage (0–100)

Using stratified sampling reduced bias and increased representation across grade levels.

III. Data Display and Exploratory Analysis

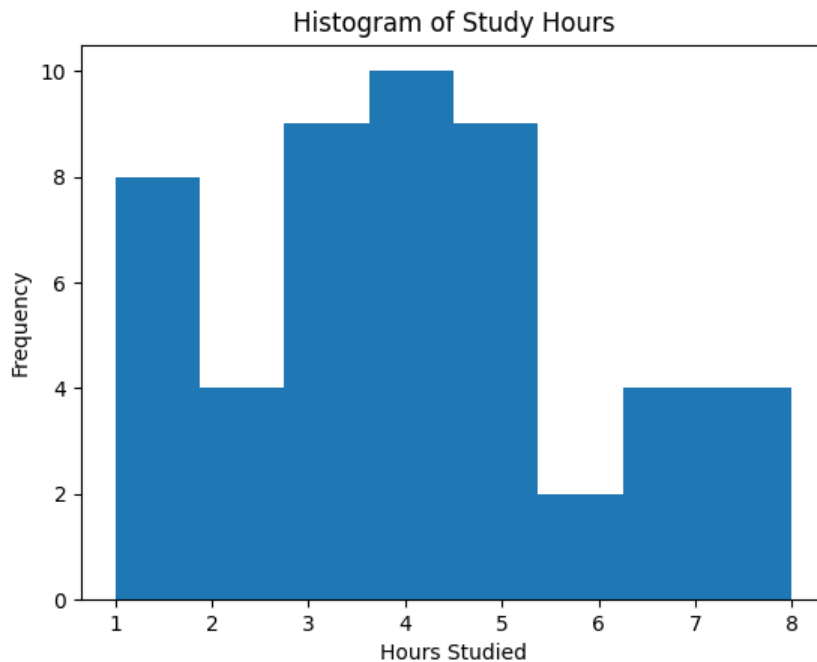


Figure 1. Histogram displaying distribution of study hours.

Histogram Interpretation: Histograms provide insight into the shape, center, spread, and possible skewness of a distribution. The distribution of study hours appears concentrated around four to six hours, with most students clustered near the five-hour benchmark. There are no major gaps and only mild variation, suggesting study time remains relatively centered.

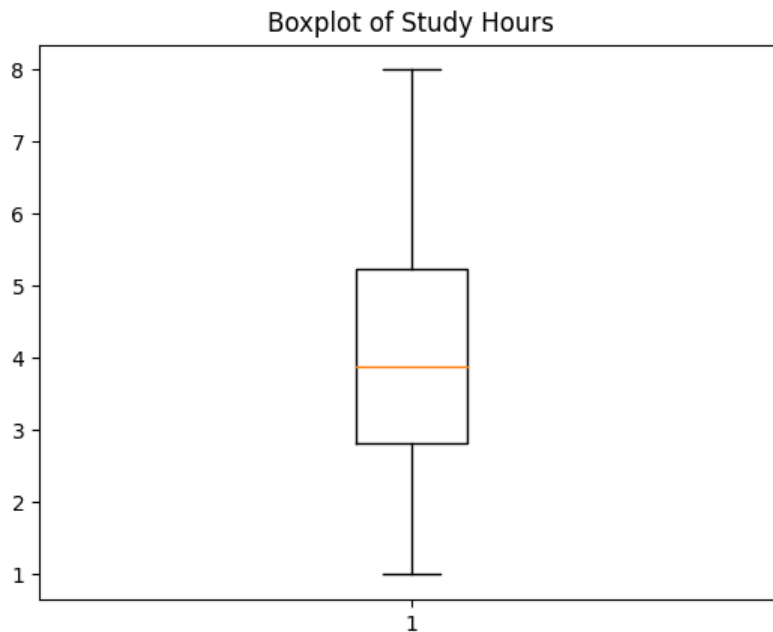


Figure 2. Boxplot of study hour distribution.

Boxplot Interpretation: The boxplot summarizes the five-number summary. The median is approximately five hours. The interquartile range suggests moderate variability in student study habits. No extreme outliers appear. The relatively symmetric appearance supports assumptions needed for later inference procedures.

IV. Inferential Analysis

Descriptive Statistics:

Study Hours:

Mean=4.58

Standard Deviation=2.06

Five Number Summary: Min=1 Q1=3 Median=5 Q3=7 Max=8

Test Scores:

Mean=83.24%

Standard deviation=5.62%

Conditions for Inference:

Random: Stratified random sampling was used.

Independence: Sample represented less than 10% of Doral Academy population.

Normal/Large Sample: Sample size $n=50$ satisfies Central Limit Theorem assumptions.

One-Sample t-Test:

$p = .155 > .05$. Because p -value exceeds $\alpha = .05$, we fail to reject the null hypothesis. There is insufficient statistical evidence suggesting average study time differs from five hours.

Two-Sample t-Test:

Students studying under five hours were compared to students studying five or more hours. $p = .0000000000046 < .05$. We reject H_0 and conclude students studying five or more hours performed significantly better.

Regression Analysis:

Predicted Score = $71.501 + 2.67(\text{Hours})$

$R^2 = .69$

Interpretation:

Approximately 69% of variation in test scores can be explained by study hours. For each additional study hour, predicted test scores increase approximately 2.67 percentage points.

V. Reflection and Future Research

Several limitations may affect interpretation of results. First, study hours and test scores were self-reported. Students may unintentionally overestimate study time or selectively report scores. Second, tests differed across classes and subjects. Variability in assessment difficulty may affect comparisons.

Motivation was also not measured. Highly motivated students may naturally study longer and perform better. Consequently, motivation could act as a confounding variable. Sleep quality, stress levels, attendance, and classroom participation were also not controlled.

Future research should examine study quality rather than study quantity alone. Variables such as active recall, note-taking strategies, distractions, and sleep patterns should be incorporated. Larger multi-school samples and longitudinal studies may improve generalizability.

VI. References (APA 7th)

Credé, M., & Kuncel, N. R. (2008). Study habits, skills, and attitudes. *Perspectives on Psychological Science*, 3(6), 425–453. <https://doi.org/10.1111/j.1745-6924.2008.00089.x>

Nonis, S. A., & Hudson, G. I. (2010). Performance of college students. *Journal of Education for Business*, 85(4), 229–238. <https://doi.org/10.1080/08832320903449550>

Plant, E. A., Ericsson, K. A., Hill, L., & Asberg, K. (2005). Why study time does not predict GPA. *Contemporary Educational Psychology*, 30(1), 96–116. <https://doi.org/10.1016/j.cedpsych.2004.06.001>